Composite of sulfur permeated in PEG polymer as the cathode of Lithium- Sulfur batteries

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***Abstract*— Development of high energy density rechargeable batteries is becoming more and more important because of the increasing energy consumption of portable and transport applications. Lithium-sulfur batteries are receiving more attention because of its highest specific-energy of 2600Wh/kg and theoretical capacity of 1672mAh/g. In addition to the high capacity, elemental sulfur also has advantages of natural abundance, low cost, environmentally benignity and low toxicity which are all the important factors for next generation of lithium batteries. The positive electrodes with polymer coating agent have been successfully prepared by a simple solution processing method. The composite has been characterized by XRD, FTIR, RAMAN, TEM and SEM-EDX analyses. The powder XRD pattern of pristine sulfur exhibited orthorhombic structure with F_{ddd} space group. In RAMAN spectra, the sulfur disappeared in the coating material, suggesting that the sulfur was enclosed into the pores of carbon black and PEG composites. Therefore, we perceive a simple strategy to improve the composite with desirable properties and better performing Li-S cells.**

Keywords—Lithium-sulfur, Carbon black, PEG, polymer.

I. INTRODUCTION

Increasing demands for the energy consumption of various portable devices such as laptops, smart grids and electric vehicle etc., batteries are looked for alternative energy storage systems for the next generation. [1,2] In view of that Lithium sulfur batteries have fascinated the greatest attention as one of the most promising candidates for the advanced energy storage systems because of its high theoretical specific capacity (1675 $mAhg^{-1}$) and theoretical energy density (2600Wh kg-1).[3,4,5]. Moreover, sulfur is abundant in nature, Low cost, Non-toxicity and environment friendly and wide temperature range of operation. Besides these, it has some disadvantages of volume expansion while charging/discharging process, is an electrical insulator, low utilization of sulfur and capacity fading. In addition to that the intrinsically low electronic and ionic conductivities of elemental sulfur and its final discharge products weaken the reversibility of electrochemical reactions on the cathode [6, 7]. Moreover, the soluble long-chain polysulfide species cause a 'chemical short' of the electrochemical cell through the 'polysulfide shuttle'. This phenomenon is accountable for high self-discharge, low columbic efficiency, severe sulfur migration, and fast capacity decay. The hail of the sulfide species outside the cathode region results in the loss of active materials and the structural failure of the cell [8-10]. Among these major problems Li-S batteries are not yet successfully commercial.

In order to inhibit polysulfides shuttling in organic electrolyte, various tactics have been suggested by research teams over the latest three decades. One of the most effective approaches is to confine sulfur into porous frameworks, such as porous carbon materials [11-13], metal oxide matrix and polymer matrix [14, 15], Graphene. The porous networks can significantly progress the capacities of the cathodes by limiting the polysulfide diffusion, while the shuttle effect has not been fully addressed. This present work is aimed at understanding the most important limiting parameters of a Li/S cell. The cathode electronic conductivity could be enhanced by preparing a composite of S and conductive materials, such as carbon and conducting polymers. Those conductive materials should preferably have high specific surface area and porous structure in order to absorb and contain polysulfides. Moreover, the structure flexibility is an important to accommodate large volume change during charge–discharge. Active carbon has been planned to use for this purpose.

In this work, the commercial conductive carbon black (CCB) and PEG which can be used as the matrix for electrode-active materials [16, 17]. Subsequently, the sulfur is loaded on the host activated CCB, PEG by heating the mixture of the sublimed sulfur and A-CCB and PEG, to produce S/A-CCB, S/A-CCB/PEG composites. As anticipated, the as-prepared composite presents the optimized electrochemical performance. Polymer modification of the carbon surface facilitates a more complete reaction by providing a chemical gradient that retards diffusion of the polysulfides into the electrolyte.

The mixed sample can be subjected to structural analysis before and after ball milling. To the samples, polymers like PEG can be added and the resulting sample can be subjected to structural studies.

II. Sample Preparation

Sulfur (Alfa Aesar) and conductive carbon black powder were weighed separately in a weight ratio of 1:3 prior to the preparation of sulfur/carbon composites. In the first step, sulfur powder was dissolved in Dimethyl Sulfoxide (DMSO) by a mantle with the temperature limit of 90˚C. Conductive carbon black powder was added to the solution and the mixture was stirred for an hour and cooled to room temperature. After that, composites were washed with ethanol and distilled water several times to eliminate the residual DMSO, then dried in a vacuum oven to evaporate moisture. The composite preparation was followed by coating process. The coating material was dissolved by water and ethanol in the volume ratio of 1:1 and the SCB was added to the above solution. The mixture was stirred mildly for an hour and allowed to precipitate gradually. Then the mixture was centrifuged and rimed several times with ethanol and dried in vacuum oven at 100˚C for 10 hours. The prepared sample was characterized by XRD, FTIR, SEM/EDX, and MICRORAMAN.

III. Results and Discussion

A. *X-Ray diffractometer*

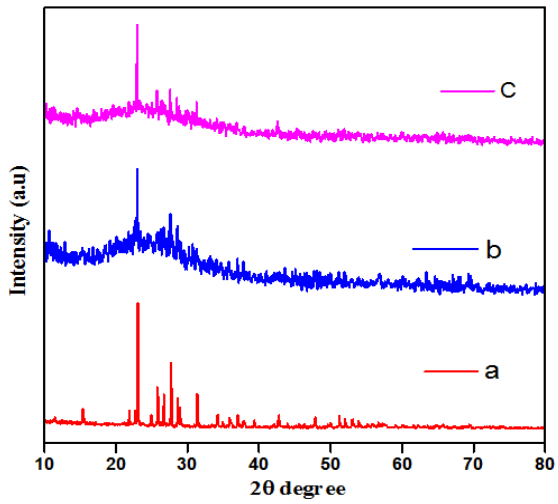

Fig.1 X-ray diffraction patterns of (a) Pristine S, (b) SCB after ball milling, (c) SCB/PEG

The crystal structures of the as-prepared samples were determined by XRD analysis. Fig.1 shows X-ray diffraction (XRD) patterns of the elemental sulfur, SCB after ball milling and SCB/PEG. The X-Ray diffraction patterns of sulfur, carbon matrix prepared from solution process method are shown in Fig.1.The reflection of the raw sulfur was indexed to an orthorhombic structure. The diffraction peaks of all the samples match very well with the standard diffraction peaks of sulfur (JCPDS 77-0145), which can be indexed to an orthorhombic phase with space group F_{ddd} [18]. Two prominent peaks exited there at 2θ=23.3° and 27°. This indicates that no phase transformation of sulfur occurs during the polyethylene glycol (PEG) coating process. In comparison with sulfur, the composites show sharp peaks of sulfur with reduced peak intensity, resulting that the sulfur are well dispersed in the coating materials.[19].

B. *Fourier Transform Infrared Spectroscopy*

Fourier Transform Infrared Spectroscopy (FTIR) offers quantitative and qualitative analysis for organic and inorganic samples. FTIR identifies chemical bonds in a molecule by producing an infrared absorption spectrum. FTIR is an effective analytical instrument for detecting functional groups and characterizing covalent bonding information. The peak at 466 cm^{-1} assigned to the stretching vibration of S–S bonds. The position of the peaks at 1196 cm^{-1} and 1103 cm^{-1} corresponding to C-O stretching and the peak located at 1634 cm^{-1} indicates the C=O stretching vibration. Peak appeared at 1350 cm^{-1} ascribed to S=O asymmetric stretching vibration and the C–S stretching vibration at 740 cm^{-1}. This analysis elucidates the functional vibrational groups in the sulfur based electrode materials.

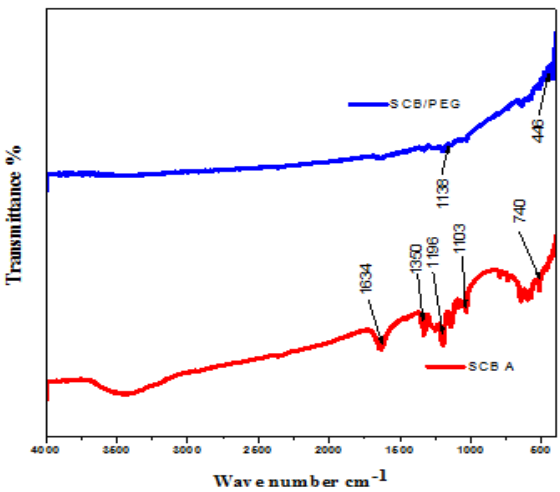

Fig 2 FT-IR Spectra of (a) SCB after ball milling, (b) SCB/PEG

C. *Micro RAMAN studies*

Raman spectra of sulfur shows a series of typical Raman peaks at 218, 242, 456 cm^{-1} corresponding to A1, E3 modes of S–S bond [18]. However, such typical sulfur peaks are vanished in the Raman spectra of SCB/PEG composites suggesting that sulfur is successfully enclosed into the pores of PEG composites, exhibits G band at 1571 cm^{-1} and D band at 1328 cm^{-1}, representing the graphite and disordered carbon respectively. It may be ascribed to that the interaction that occurs between the incorporated sulfur and unsaturated bonds in PEG dominant the decrease of disordered carbon and defect amount [18].

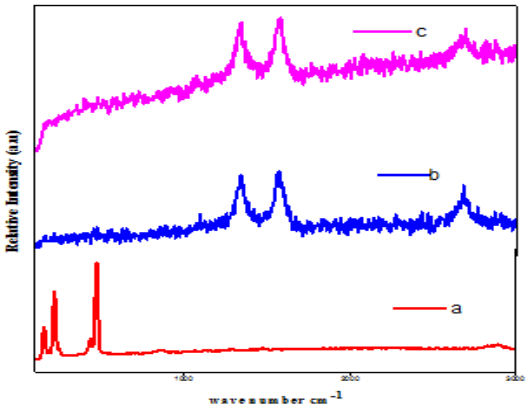

Fig. 3 Raman Spectra of (a) Pristine Sulfur, (b) SCB after ball milling,
(c) SCB/PEG composite

D. *Scanning Electron Microscopy*

The Fig.4-5 shows the SEM image of sample SCB/PEG with various magnifications. The average size of the sulfur particle was found to be less than 1µm, resulted from the size limiting effect afforded by the surfactant coating on the sulfur particles [20]. From the fig it was observed that no descry sulfur particles can be found outside of the SCB/PEG particles which offer the full embodiment of sulfur into the channel of carbon black [21]. This polymer modified composite clearly exhibited better morphological control than the unmodified composite [22]. The EDX analyses were used for the analysis of compositions of the comprising contents which are shown in Fig.6.

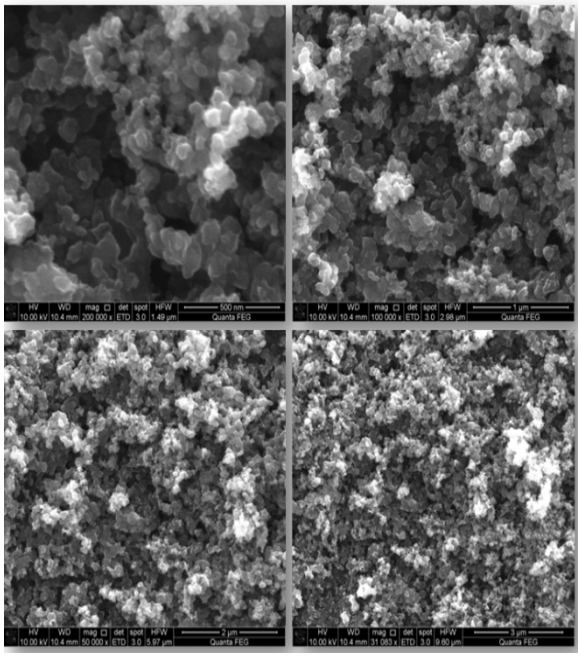

Fig. 4 HRSEM-Image of sulfur carbon material without coating

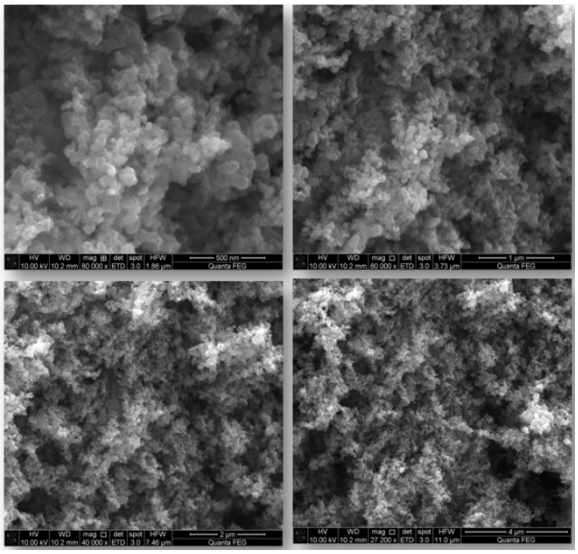

Fig .5 HRSEM-Image of Sulfur Carbon Coating with PEG

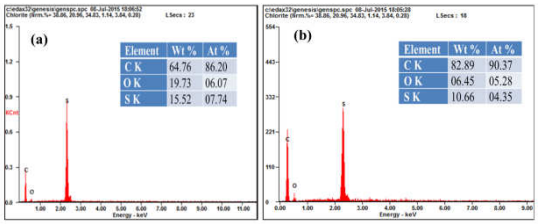

Fig. 6. EDX-Image of a) Sulfur carbon material After Ball Milling b) Sulfur Carbon Coating with PEG

D. *Transmission Electron Microscopy*

From TEM image, it was clearly found that the sulfur had encapsulated into the carbon matrix. It was observed that SCB/PEG composites possesses unique sandwich architecture in which S nanoparticles were well embedded into the chain like structure of Carbon Black and the PEG smudged the SCB composites. PEG modification of the SCB surface facilitates a more complete reaction by providing a chemical gradient that retards diffusion of the polysulfides into the electrolyte [6].

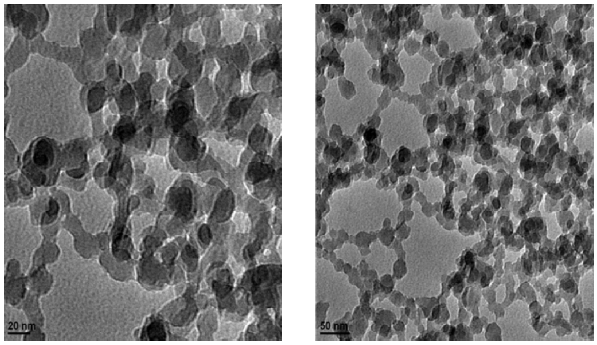

Fig .7 TEM Images of Sulfur Carbon Coating with PEG

IV. Conclusion

The positive electrodes with polymer coating agent have been successfully prepared by a simple solution processing method. The mixture of SCBA, SCB/PEG has been characterized by XRD, FTIR, RAMAN, TEM and SEM-EDX analyses. The powder

XRD pattern of pristine sulfur exhibited orthorhombic structure with F_{ddd} space group. The diffraction peaks with reduced intensity indicate that the sulfur was fully incorporated into the pores of coating material.

The SEM images show that the average size of the sulfur/CCB particle was found to be less than 1μm. In RAMAN spectra, the sulfur disappeared in the coating material, suggesting that the sulfur was enclosed into the pores of carbon black and PEG composites. Thus it is concluded that the positive electrode of sulfur was prepared successfully. The coating material overruling in the dissolution of polysulfides and may enhance the conductivity of Li-S batteries. Therefore, we perceive a simple strategy to improve the composite with desirable properties and better performing Li-S cells can be expected for the application of high rate energy storage in the future.

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